

Real-life intrusion Analysis - Michael Alpin

Student Name

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Chosen Cyber Technique

MitM Evil Twin attack

Selected Cyber Attack/Campaign

The coffee shop WiFi incident:

<https://www.studiodsharp.com/blog/coffee-shop-wifi-incident>

Case Overview

Attribute	Details
Attacker / Threat Actor	A large (anonymous) operation of fake baristas/workers targeting coffee shops
Victim / Target Organization	Regular coffee customers
Attack Timeline	The attack happened continuously throughout a long period (no year listed)
Overall Impact & Damage	A lot of the coffee shops customers suffered from getting their data stolen and having their privacy violated

Technical Attack Architecture & Technique Execution

Initial Access Vector

The easy access to the coffee shops made the penetration mission for the attackers very easy. The attackers would go to work in the coffee shops, and then proceed to secretly install in there the equipment running the attack.

Mechanics of the Chosen Technique & Attack Flowchart/Timeline

This is how the attack was executed:

- The attackers would open the fake WiFi hotspots, impersonating to be the legitimate WiFi of the place

And after unsuspecting customers would join, the attackers would then proceed to:

- Log all HTTP traffic
- Attempt SSL downgrade attacks (forcing HTTPS to HTTP)
- Inject ads
- Capture login attempts

All of this by running Hostapd, DNSMasq, SSLstrip, Ettercap and other custom scripts.

Using those mechanics, the attackers would gain money, a database of the customers's browsings, the private information of the customers and their login credentials.

Detection Analysis

How it Was Detected

Luckily for the shop and the other customers, one of the customers (the author) happened to work in cybersecurity. Here is how he detected the attack:

- The author noticed that only a few minutes into the connection, his connection started to drop, and the device speed slowed
- After starting to get suspicious, the author proceeded to run netstat, arp and nmap requests, noticing suspicious results (weird getaway IP, a big amount of devices on the network etc.)
- Then he performed WiFi scan, seeing there are multiple WiFi networks running in the place, each one having a slightly different name
- After that, the author has opened Wireshark to sniff the network traffic, confirming his suspicion by noticing the traffic being logged and noticing the attacker's scripts

How it Could Have Been Detected

Here is how the attack could've been detected ahead of the damage:

- Regular WiFi network scan: scanning the open WiFi networks regularly could've give an early detection of the attack
- Regular network traffic logs: performing network logs regularly and checking for suspicious behavior can prevent attackers from penetrating into the network, and running malicious scripts

Key Takeaways & Mitigation

Primary Mitigation Strategy

To neutralize this attack technique, a script running regularly checking all of the open networks in the area, scanning both the networks and their metadata, and the traffic in them, and then dealing with networks he detects to be malicious, could neutralize this attack.

Lessons for Intrusion Detection

As this attack happened in a regular coffee shop, targeting innocent customers, and relied on the unsuspecting of the shop owners to take place, we can learn that one of the biggest strengths of this attack is how easy and accessible it is to be performed, although it may easily be detected.

As of our model detection, I am taking from this real-life case the importance of scanning for the suspicious data harvesting methods.

References

Dennis Sharp, "The Coffee Shop WiFi Incident: A Security Lesson",
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